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Macros

Macros may be the single most important reason why Lispers put up with all those annoying parentheses in their code. These very parentheses enable this powerful macro system in Lisp. Paul Graham, who comes close to being a 'Lisp missionary', points out that Lisp code is made out of Lisp data objects. And not in the trivial sense that the source files contain characters, and strings are one of the data types supported by the language. Lisp code, after it's read by the parser, is made of data structures that you can traverse.

If you understand how compilers work, you'll figure out that what's really going on is not so much that Lisp has a strange syntax (parentheses everywhere!) as that Lisp has no syntax. You write programs in the parse trees that get generated within the compiler when other languages are parsed—but these parse trees are fully accessible to your programs; you can write programs that manipulate them. In Lisp, these programs are called macros. They are programs that write programs.

A great medium to express recursion

Recursion is the act of defining an object or solving a problem in terms of itself. Properly used, recursion is a powerful problem-solving technique, both in artificial domains like mathematics and computer programming, as well as in real life.

"The power of recursion evidently lies in the possibility of defining an infinite set

of objects by a finite statement. In the same manner, an infinite number of computations can be described by a finite recursive program, even if this program contains no explicit repetitions," wrote Niklaus Wirth in his 1976 book, 'Algorithms + Data Structures = Programs' (see 'References').

Lisp is the best programming language to use when working with recursive problems. Daniel P. Friedman and Matthias Felleisen demonstrate this case for Lisp in their book 'The Little Lisper'. Lisp is inherently symbolic—programmers do not have to make an explicit mapping between the symbols of their own language and the representations in the computer. Recursion is Lisp's natural computational mechanism; the primary programming activity is the creation of (potentially) recursive definitions.

What else?

There are many more reasons why one should try out Lisp. Apart from the ones mentioned above, I have covered a few in my previous articles. This series was my small attempt to evangelise Lisp and plant a seed of curiosity in those who are seeking a better way of life in the maddening programming world. If the seed has taken root in your mind and you'd like to explore further, feel free to contact me at vivek@codeCraft.in. In, should you have any questions, or if you need pointers to resources that may aid your discovery of the world's most powerful programming language.

Have fun with Lisp! 

References

- [1] Wirth, Niklaus (1976), 'Algorithms + Data Structures = Programs', Prentice-Hall.

By: Vivek Shangari

The author is a hard-core hacker, if ever there was one. He always thinks programs, and holds a piece of code in his head all the time. His favourite past time is to stop random strangers on the street and start talking about the benefits of open source over proprietary software, till they agree to switch sides, or threaten to jump off the nearest building. Every now and then, he manages to talk a bunch of unsuspecting potential programmers into letting him train them in Lisp. It's not rare to find people stumble out after the training session 'zombified', or with tears running down their faces. "Those are tears of joy!" he will patiently explain. For those of you out there who cannot be deterred, you can reach him at vivek@codeCraft.in.